

WanderWise

Smart Travel Planner

Capstone Project Report

Domain: Travel & Tourism

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1. Project Overview

WanderWise is a full-stack React web application for the Travel & Tourism domain that allows users to plan, manage and analyse their travel trips. The application provides an end-to-end solution: from discovering destinations and viewing live weather, to creating detailed trips with itineraries and expenses, and analysing spending via interactive charts.

Problem Statement

Travellers struggle with fragmented tools for trip planning, budget tracking and destination research. WanderWise consolidates all of these into one cohesive React SPA, integrating free public APIs for real-time data and using Redux Toolkit for scalable state management.

Unique Aspects

- Domain: Travel & Tourism (India-focused, 12 curated destinations)
- APIs: Open-Meteo (live weather) + REST Countries (country metadata) – both free, no API key needed
- Feature Set: 9 advanced features implemented (Authentication, Pagination, Search+Filter+Sort, Dark Mode, Debouncing, Error Boundary, Lazy Loading, Charts, Multi-step Form)

2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite	Component-based SPA, fast dev server
Language	JavaScript ES6+	Arrow fns, destructuring, async/await
State Mgmt	Redux Toolkit	Global state: trips, explore, slices
Routing	React Router v6	Nested routes, protected routes, lazy
API Client	Axios	HTTP requests with interceptors
Styling	Tailwind CSS	Utility-first, dark mode, custom tokens
Charts	Recharts	Pie, Bar, Line charts on Dashboard
Deployment	Vercel / Netlify	CI/CD from GitHub, instant preview

3. Features Implemented

3.1. Authentication & Role-based Access

Register/Login with localStorage persistence. AuthContext provides user state. ProtectedRoute guards all main pages. Logout clears session.

3.2. Trip CRUD Operations

Create (multi-step form), Read (list + detail), Update (expense add/delete), Delete (with confirmation). Redux Toolkit manages all state with selectors.

3.3. Expense Tracker

Per-trip expense management with 6 categories. Category breakdown bar chart. Real-time budget vs spent progress bar.

3.4. Explore Destinations

12 curated Indian destinations. Click any card to open a modal with live weather (Open-Meteo) and country info (REST Countries API).

3.5. Dashboard with Charts

4 chart types: Status pie, Category pie, Budget vs Spent bar, Monthly spend line chart – all via Recharts.

3.6. Search + Filter + Sort

Debounced search (400ms), status/region filter, sort by date/budget/name – on both Trips and Explore pages.

3.7. Dark Mode Toggle

CSS custom properties + Tailwind dark class. ThemeContext persists preference to localStorage.

3.8. Debounced API Calls

Custom useDebounce(value, 400) hook prevents excessive API calls and Redux dispatches while typing.

3.9. Pagination

Explore page shows 6 destinations per page with a full Pagination component. State lives in Redux.

3.10. Error Boundary

Class-based ErrorBoundary wraps all routes. Catches render errors and shows a friendly fallback UI.

3.11. Lazy Loading

All 9 page components are React.lazy + Suspense loaded. Reduces initial bundle size significantly.

3.12. Multi-step Form

3-step NewTrip form with per-step validation, progress indicators, emoji picker, and tag selection.

4. API Integration

Open-Meteo API (<https://api.open-meteo.com>)

Free, no API key. Returns current temperature, weather code, and wind speed. Used in DestinationModal when a user clicks a destination card. Fetched via createAsyncThunk in exploreSlice.

REST Countries API (<https://restcountries.com>)

Free, no API key. Returns country name, capital, population, flag SVG, subregion, and timezone. Fetched alongside weather in the same modal interaction.

Both APIs are fetched only on user interaction (lazy), with loading skeletons and error states handled gracefully.

5. Project Structure

```
src/
  ├── components/ - Reusable UI (Layout, TripCard, DestinationCard,
  │               SearchBar, Pagination, ErrorBoundary)
  ├── pages/ - 9 lazy-loaded route pages (Home, Explore, Trips, TripDetail,
  │   NewTrip, Dashboard, Login, Register, NotFound)
  ├── store/ - Redux Toolkit (store.js, tripsSlice.js, exploreSlice.js)
  ├── context/ - React Context (AuthContext, ThemeContext)
  ├── hooks/ - Custom hooks (useDebounce)
  └── utils/ - Helpers (formatDate, formatCurrency, genId, etc.)
```

6. Performance Optimizations

- Code splitting via React.lazy – all pages loaded on demand, reducing initial JS bundle.
- Debounced search inputs – API calls and Redux dispatches throttled to 400ms.
- Memoized selectors – selectFilteredTrips and selectPaginatedDestinations avoid redundant recalculations.
- Pagination – only 6 destination cards rendered per page instead of all 12.
- CSS transitions over JS animations – hardware-accelerated, no layout thrashing.

7. Deployment

The application is deployed to Vercel / Netlify from a GitHub repository.

GitHub Repository: [https://github.com/\[username\]/wanderwise](https://github.com/[username]/wanderwise)

Live Demo: <https://wanderwise.vercel.app>

8. Submission Checklist

Item	Status
GitHub repository (unique)	Done
Live deployed link	Done
Project report (PDF)	Done
Demo video (optional)	Pending
Viva / presentation ready	Ready
Anti-copy: unique domain + API + feature set	Confirmed